Coordinate-free Mathematics

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Contents

The first section develops algebraic constructions independent of a basis; the second develops geometric constructions independent of coordinates; the third develops analytic constructions from norms, operators, and positivity.

1 Algebra

Linear algebra studies linear maps between vector spaces. A **vector space** V over a field k is a set equipped with vector addition and scalar multiplication operations satisfying straightforward compatibility conditions.

A **linear map** $f : V \to U$ between k-vector spaces is a map of underlying sets such that $f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$ for all $\alpha, \beta \in k$ and $v, w \in V$.

Exercise 1 (Small string vibrations obey superposition)

Derive the wave equation starting from a taut string and show that its solutions form a vector space. Show that time evolution from initial data to the state at time t is a linear map.

The **free k-vector space on a set** X, denoted kX, is the set of finitely supported set-theoretical functions $X \to k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W, denoted V/W, is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is the vector space $k(V \times W)/N$ where N is the subspace generated by the obvious linearity relations on V and W.

An **isomorphism** of vector spaces $V \cong W$ consists of linear maps $f : V \to W$ and $g : W \to V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$.

Exercise 2 (Universal constructions)

Give the universal properties of free spaces, direct sums, quotients, and tensor products, and use them to prove the three isomorphism theorems and $k(X \sqcup Y) \cong kX \oplus kY$, $k(X \times Y) \cong kX \otimes kY$.

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k. The evaluation map $\mathrm{ev}_V : V^* \otimes V \to k$ is the linear extension of the evaluation morphism $f \otimes x \mapsto f(x)$. We say that V is **finite dimensional** if there is a co-evaluation morphism $\mathrm{coev}_V : k \to V \otimes V^*$ satisfying the snake equations $(1_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes 1_V) = 1_V$ and $(\mathrm{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \mathrm{coev}_V) = 1_{V^*}$. We define the **dimension** $\dim(V) = \mathrm{tr}(1_V)$, where $\mathrm{tr}(f) : \mathrm{End}(V) \to k$ is given by $f \mapsto (\mathrm{ev}_V \circ s_{V,V^*} \circ (f \otimes 1_{V^*}) \circ \mathrm{coev}_V)(1)$ where $s_{X,Y} : X \otimes Y \to Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$.

Exercise 3 (Trace and dimension identities)

Prove the following for finite-dimensional vector spaces V, W and linear maps $f, g : V \to V$.

- $\mathrm{tr}(fg) = \mathrm{tr}(gf)$

- $\operatorname{tr}(f \oplus g) = \operatorname{tr}(f) + \operatorname{tr}(g)$
- $\operatorname{tr}(f \otimes g) = \operatorname{tr}(f)\operatorname{tr}(g)$
- $\dim(V) \in \mathbf{N}$
- $V \cong W$ *iff* $\dim(V) = \dim(W)$

Define the tensor algebra and **exterior algebra of** V by

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \qquad \Lambda V = T(V)/\langle v \otimes v : v \in V \rangle.$$

The space $\Lambda^k V$ is the degree-k component of ΛV.

Exercise 4 (The top exterior power is one-dimensional) *Show that the highest graded component of ΛV is one-dimensional.*

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by $v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots f(v_{\dim(V)})$.

Exercise 5 (Determinant identities and volume)
Prove the following:

- $\det(f \oplus g) = \det(f)\det(g)$
- *If $V = \mathbb{R}^n$, then $\det(f)$ is the signed volume of $f(I^n)$ where $I = [0, 1]$.*
- *Cramer's rule*

For $f \in \operatorname{End}(V)$, the **characteristic polynomial** is the polynomial $\lambda \mapsto \det(f - \lambda 1_V)$.

A scalar $\lambda \in k$ is an **eigenvalue** of f if $\ker(f - \lambda 1_V) \neq 0$. This kernel is its **eigenspace**; its dimension is the geometric multiplicity of λ.

Exercise 6 (Characteristic polynomial and Jordan form)
Let $f \in \operatorname{End}(V)$ have split characteristic polynomial. Prove the Cayley–Hamilton theorem and Jordan normal form, and derive the algebraic and geometric multiplicity inequality together with the eigenvalue formulas for $\operatorname{tr}(f)$ and $\det(f)$.

Exercise 7 (Critical damping is a Jordan block)
Use Jordan normal form to solve $m\ddot{x} + b\dot{x} + \kappa x = 0$ in all three damping regimes, and show that critical damping is exactly $b^2 = 4m\kappa$, where a nontrivial Jordan block produces $te^{-bt/(2m)}$.

A **Hermitian inner product** on a complex vector space V is a positive-definite sesquilinear form $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable and linear in the second. It induces the norm $\|v\| = \sqrt{\langle v, v \rangle}$. For a linear map $A : V \to W$

between finite-dimensional Hermitian spaces, an **adjoint** is a map $A^* : W \to V$ satisfying
$$\langle Av, w\rangle = \langle v, A^* w\rangle.$$

An endomorphism A is **self-adjoint** if $A = A^*$, **normal** if $A^*A = AA^*$, **positive** if $\langle v, Av\rangle \geq 0$ for every v, and **unitary** if $A^*A = AA^* = 1$. Equip $\mathrm{End}(V)$ with the induced operator norm.

Exercise 8 (Finite-dimensional spectral theorem)
Develop the finite-dimensional spectral theorem: prove existence of adjoints and positive square roots, and prove that every normal operator has an orthogonal spectral decomposition

$$A = \sum_\lambda \lambda p_\lambda, \qquad \sum_\lambda p_\lambda = 1.$$

Deduce the singular-value decomposition

$$T = U\Sigma V^*.$$

Exercise 9 (The spectral decomposition produces normal modes)
For a fixed-end chain of N equal masses and springs, set $q_0 = q_{N+1} = 0$ and define
$$(Kq)_j = \frac{\kappa}{m}(2q_j - q_{j-1} - q_{j+1}).$$

Find the spectral decomposition of K and use it to determine all normal modes of $\ddot{q} + Kq = 0$, their frequencies, and the conserved energy

$$\mathcal{E}(q, \dot{q}) = \frac{m}{2}\|\dot{q}\|^2 + \frac{m}{2}\langle q, Kq\rangle.$$

Exercise 10 (Fourier modes solve the fixed-end wave equation)
Solve the fixed-end wave equation from Exercise 1 by Fourier series, prove conservation of wave energy, and recover its normal-mode frequencies as the continuum limit of Exercise 9.

1.1 Group representations

An action of a group G on a set X assigns to each $g \in G$ a bijection $x \mapsto gx$ such that $1x = x$ and $(gh)x = g(hx)$. A **representation** of G on a vector space V is a homomorphism
$$\rho : G \longrightarrow \mathrm{GL}(V).$$

A subspace $W \subseteq V$ is **G-invariant** if $\rho(g)W \subseteq W$ for every $g \in G$. Restriction gives the **subrepresentation** W, and $\rho(g)(v + W) = \rho(g)v + W$ gives the **quotient representation** V/W. A nonzero representation is **irreducible** if its only invariant subspaces are 0 and V.

For representations V and W, the direct sum, tensor product, and dual representations are defined by

$$\begin{aligned} g(v, w) &= (gv, gw), \\ g(v \otimes w) &= gv \otimes gw, \\ (g\phi)(v) &= \phi(g^{-1}v), \qquad \phi \in V^*. \end{aligned}$$

An **intertwiner** $T : V \to W$ is a linear map satisfying $T\rho_V(g) = \rho_W(g)T$ for every g. Write $\mathrm{Hom}_G(V, W)$ for the space of intertwiners and

$$V^G = \{v \in V : \rho(g)v = v \text{ for every } g \in G\}$$

for the fixed-point space.

Exercise 11 (Averaging produces invariant projections and inner products)

Let G be finite and let V be a complex representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

Prove that P_G is the projection of V onto V^G. Prove also that averaging any Hermitian inner product over G produces a G-invariant Hermitian inner product, so every finite-dimensional complex representation of G is equivalent to a unitary representation.

1.2 Maschke's theorem and Schur's lemma

Exercise 12 (Maschke's theorem)

Let $W \subseteq V$ be G-invariant and let $P : V \to W$ be any linear projection. Define

$$\widetilde{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g)^{-1}.$$

Prove that $\widetilde{P}$ is G-equivariant and has image W. Deduce that $\ker \widetilde{P}$ is a G-invariant complement of W. If G is finite and the characteristic of k does not divide $|G|$, prove that every finite-dimensional k-representation of G is a direct sum of irreducible representations.

Exercise 13 (Schur's lemma)

Let V and W be irreducible finite-dimensional complex representations. Prove that every nonzero intertwiner $T : V \to W$ is an isomorphism and that

$$\mathrm{End}_G(V) = \mathbb{C}1_V.$$

Let $\mathbb{C}[G]$ act on V by the linear extension of ρ. Deduce that every element of the center $Z(\mathbb{C}[G])$ acts by a scalar on each irreducible representation.

1.3 Characters and orthogonality

The **character** of a finite-dimensional representation V is the function

$$\chi_V(g) = \mathrm{tr}(\rho_V(g)).$$

For class functions $\chi, \psi : G \to \mathbb{C}$, define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

Exercise 14 (Great Orthogonality Theorem)
For irreducible unitary representations Γ^α and Γ^β of dimensions d_α, d_β of a finite group G, prove

$$\sum_{g \in G} \Gamma^\alpha(g)_{ij} \overline{\Gamma^\beta(g)_{k\ell}} = \frac{|G|}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{j\ell}.$$

Develop its character consequences, including

$$m_\alpha = \langle \chi_\alpha, \chi_V \rangle_G.$$

Deduce that characters classify finite-dimensional complex representations.

For an irreducible character χ_α, define the **symmetry projection**

$$P_\alpha = \frac{d_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho(g).$$

Exercise 15 (Symmetry projections)
Prove that the operators P_α are pairwise orthogonal idempotents and that $P_\alpha V$ is the α-isotypic component of V. Interpret P_α as constructing symmetry-adapted linear combinations of atomic orbitals or displacement vectors.

1.4 Molecular symmetry

The **point group** of a molecule is the finite group $G \subseteq O(3)$ of spatial symmetries preserving its nuclear configuration, including atomic species. For a molecule with N atoms, G acts on the real displacement space

$$V_{\mathrm{disp}} = \bigoplus_{a=1}^{N} \mathbb{R}^3_a \cong \mathbb{R}^{3N}$$

by permuting atoms and applying the spatial linear action to each displacement. The three-dimensional translational representation carries the polar-vector action R_g, while the rotational representation carries the axial-vector action $\det(R_g) R_g$; denote them by V_{trans} and V_{rot}. We complexify these real representations when applying character theory. For a nonlinear molecule, the vibrational representation is the class

$$[V_{\mathrm{vib}}] = [V_{\mathrm{disp}}] - [V_{\mathrm{trans}}] - [V_{\mathrm{rot}}]$$

in the representation ring.

Exercise 16 (Character of molecular displacements)
For $g \in G$, prove that

$$\chi_{\mathrm{disp}}(g) = \sum_{\substack{a \\ g(a)=a}} \mathrm{tr}(R_g),$$

where $R_g : \mathbb{R}^3 \to \mathbb{R}^3$ is the spatial action and the sum is over atoms fixed by g. Prove that $\dim V_{\mathrm{vib}} = 3N - 6$ *for a nonlinear molecule, and use character orthogonality to express the multiplicity of every irreducible representation in V_{vib}.*

For water, take the z-axis along the C_2 axis and the molecule in the yz-plane. Its point group is

$$C_{2v} = \{E, C_2, \sigma_{xz}, \sigma_{yz}\}.$$

Exercise 17 (Water has three symmetry-classified vibrations)
Derive the C_{2v} character table and use it to show that water's vibrational representation is

$$V_{\mathrm{vib}} \cong 2A_1 \oplus B_2.$$

Identify and construct the three normal modes.

1.5 Spectroscopic selection rules

Let V_i and V_f be the symmetry types of initial and final states. The electric-dipole operator transforms as the vector representation $V_\mu = V_{\mathrm{trans}}$. A transition amplitude

$$\langle \psi_f, \mu\psi_i \rangle$$

is the matrix coefficient used below.

Exercise 18 (Symmetry determines electric-dipole transitions)
Derive the electric-dipole selection rule

$$1 \subseteq V_f^* \otimes V_\mu \otimes V_i.$$

Apply it to show that all three fundamental vibrations of water are infrared active.

Exercise 19 (All fundamental vibrations of water are Raman active)

Let the polarizability tensor carry $\mathrm{Sym}^2(V_\mu)$*. Prove that a vibrational mode is Raman active precisely when its irreducible representation occurs in this symmetric square. Prove that*

$$\mathrm{Sym}^2(V_\mu) \cong 3A_1 \oplus A_2 \oplus B_1 \oplus B_2$$

for C_{2v}*. Deduce that all three fundamental vibrational modes of water are Raman active.*

1.6 Lie groups, infinitesimal symmetry, and spin

The molecular point groups above are discrete subgroups of $O(3)$. Define the group of orientation-preserving spatial rotations by

$$\mathrm{SO}(3) = \{R \in M_3(\mathbb{R}) : R^T R = 1,\ \det R = 1\}.$$

A **matrix Lie group** is a subgroup $G \subseteq \mathrm{GL}(n,\mathbb{C})$ that is closed in the subspace topology. Its **Lie algebra** is

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) : e^{tX} \in G \text{ for all } t \in \mathbb{R}\},$$

where $e^{tX} = \sum_{k\geq 0}(tX)^k/k!$. The Lie bracket is $[X,Y] = XY - YX$. An abstract **Lie algebra** over k is a k-vector space with a bilinear bracket that is skew-symmetric and satisfies the Jacobi identity. A **Lie algebra representation** on V is a linear map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ with $\rho([X,Y]) = [\rho(X),\rho(Y)]$.

For $g \in G$ and $X, Y \in \mathfrak{g}$, the adjoint actions are

$$\mathrm{Ad}_g(X) = gXg^{-1}, \qquad \mathrm{ad}_X(Y) = [X,Y].$$

Exercise 20 (Classical matrix Lie algebras)

Prove that the Lie algebras of $\mathrm{GL}(n,\mathbb{C})$*,* $\mathrm{SL}(n,\mathbb{C})$*,* $\mathrm{O}(n)$*,* $\mathrm{SO}(n)$*,* $\mathrm{U}(n)$*, and* $\mathrm{SU}(n)$ *are respectively*

$$\mathfrak{gl}(n,\mathbb{C}), \quad \mathfrak{sl}(n,\mathbb{C}) = \{X : \mathrm{tr}(X) = 0\},$$

$$\mathfrak{o}(n) = \{X : X^T = -X\}, \quad \mathfrak{u}(n) = \{X : X^* = -X\}, \quad \mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n,\mathbb{C}).$$

Verify that each is a Lie algebra under $[X,Y] = XY - YX$*.*

Exercise 21 (One-parameter subgroups and the bracket)

Prove that $t \mapsto e^{tX}$ *is a smooth group homomorphism* $\mathbb{R} \to G$ *whenever* $X \in \mathfrak{g}$*, and that every smooth homomorphism* $\mathbb{R} \to G$ *arises uniquely in this way. Prove that*

$$[X,Y] = \left.\frac{d}{dt}\right|_{t=0} \left.\frac{d}{ds}\right|_{s=0} e^{tX}e^{sY}e^{-tX}e^{-sY},$$

and deduce $\mathrm{Ad}_{e^{tX}} = e^{t\,\mathrm{ad}_X}$*.*

Exercise 22 (Differentiating group representations)

Let $\rho : G \to \mathrm{GL}(V)$ be a smooth finite-dimensional representation of a matrix Lie group. Prove that

$$\dot\rho(X) = \left.\frac{d}{dt}\right|_{t=0} \rho(e^{tX})$$

is a Lie algebra representation $\mathfrak{g} \to \mathfrak{gl}(V)$, and that $\rho(e^{tX}) = e^{t\dot\rho(X)}$. If G is connected, prove that a subspace is G-invariant if and only if it is $\mathfrak{g}$-invariant.

Exercise 23 (Rotations, angular momentum, and spin)

Using the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Identify $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, construct the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, and classify the irreducible spin representations $\rho_j : SU(2) \to U(V_j)$. For $J_a = i\dot\rho_j(-i\sigma_a/2)$, derive $\dim V_j = 2j+1$ and $J^2 = j(j+1)1_{V_j}$, including the distinction between integer and half-integer spin.

A **quantum number** is an eigenvalue used to label a joint eigenspace of commuting observables.

Exercise 24 (Energy and symmetry produce quantum numbers)

Let H be a self-adjoint operator on a finite-dimensional Hermitian space V with a commuting $\mathrm{SU}(2)$ action. Derive the joint quantum-number decomposition

$$|E, j, m, \alpha\rangle$$

for H, J^2, J_3, including the allowed values and the degeneracy forced by spin j.

Exercise 25 (Hidden symmetry determines the hydrogen spectrum)

With $p = -i\hbar\nabla$, $r = |x|$, and $L = x \times p$, consider the hydrogen Hamiltonian and Runge–Lenz operator

$$H = \frac{p^2}{2\mu} - \frac{\kappa}{r}, \qquad A = \frac{1}{2\mu}(p \times L - L \times p) - \kappa\frac{x}{r}.$$

Use the hidden $\mathfrak{so}(4)$ symmetry to derive the bound-state spectrum and degeneracies

$$E_n = -\frac{\mu\kappa^2}{2\hbar^2 n^2}, \qquad n \geq 1, \qquad \dim W_n = n^2,$$

and the angular-momentum decomposition

$$W_n \cong \bigoplus_{\ell=0}^{n-1} V_\ell.$$

Recover the quantum numbers and transition frequencies, and compare them with hydrogen spectroscopy [16] and Pauli's algebraic derivation [15].

1.7 Monoidal and braided categories

A $\mathbb{C}$-linear category is a category whose morphism sets

$$\mathrm{Hom}_{\mathcal{C}}(x, y)$$

are complex vector spaces and whose composition is bilinear.

A $\mathbb{C}$-linear monoidal category is a $\mathbb{C}$-linear category equipped with a tensor product $\otimes$ that is bilinear on morphisms and is associative with tensor unit 1, up to coherent natural isomorphism.

Let $\mathcal{C}$ be such a category with finite direct sums. An object x is **simple** if $\mathrm{End}(x) \cong \mathbb{C}$, and $\mathcal{C}$ is **semisimple** if every object is a finite direct sum of simple objects. Assume that 1 is simple, that there are finitely many isomorphism classes of simple objects, and that $\mathcal{C}$ is **rigid**: every x has a dual x^* with evaluation and coevaluation maps satisfying the snake identities.

The definitions of duality, trace, and dimension above extend verbatim to $\mathcal{C}$. Write $d_x = \mathrm{tr}(1_x)$. Assume that the categorical dimensions of nonzero objects are positive real numbers.

Exercise 26 (Categorical trace and dimension)

Repeat the first three parts of Exercise 3 in $\mathcal{C}$.

For $x, y, z \in \mathcal{C}$, write

$$F^{xyz} : (x \otimes y) \otimes z \longrightarrow x \otimes (y \otimes z)$$

for the associator. A **braiding** is a family of isomorphisms

$$c_{x,y} : x \otimes y \longrightarrow y \otimes x.$$

For $f : x \to x'$ and $g : y \to y'$, it satisfies $(g \otimes f)c_{x,y} = c_{x',y'}(f \otimes g)$. The associator satisfies the pentagon identity, and the associator and braiding satisfy the hexagon identities. Suppressing associators, the latter are

$$c_{x,y\otimes z} = (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z),$$
$$c_{x\otimes y,z} = (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}).$$

The **monodromy** of x and y is the double braiding

$$M_{x,y} = c_{y,x}c_{x,y} : x \otimes y \longrightarrow x \otimes y.$$

Thus $c_{x,y}$ is one crossing, while $M_{x,y}$ is a full braid. If $x \otimes y$ is simple, identify $M_{x,y}$ with its scalar in $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Decompositions of tensor products of simple objects are **fusion rules**, and the spaces $\mathrm{Hom}_{\mathcal{C}}(z, x \otimes y)$ are **fusion spaces**.

An object x is **invertible** if $x \otimes x^* \cong 1$.

Exercise 27 (The $\nu = 1/3$ anyonic phase advances by $2\pi/3$)

Assume that every simple object is invertible and that their isomorphism classes form $\mathbb{Z}/m$, generated by a. Classify its monodromy, writing $M_{r,s} = M_{a^r,a^s}$, and show that, for some $k \bmod m$,

$$M_{r,s} = \exp\left(\frac{2\pi i k r s}{m}\right).$$

For $k = 1$ and $m = 3$, derive the $2\pi/3$ phase advance per enclosed anyon and compare it with [27].

Exercise 28 (Topological quantum computation)

Let $\mathcal{C}$ be a semisimple rigid braided $\mathbb{C}$-linear category with positive categorical dimensions, simple objects $1, \tau$, and fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$. Set $V = \mathrm{Hom}_{\mathcal{C}}(\tau, \tau^{\otimes 3})$ and $\varphi = (1+\sqrt{5})/2$, and in the fusion basis take [26]

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \qquad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}.$$

Derive $d_\tau = \varphi$ and the Fibonacci growth of the fusion spaces. Prove that $b_1 \mapsto R$ and $b_2 \mapsto F^{-1}RF$ define a non-Abelian unitary representation $\rho : B_3 \to \mathrm{U}(V)$ whose projective image is infinite and dense in $\mathrm{PU}(V) \cong \mathrm{SO}(3)$, equivalently that it approximates every element of $\mathrm{SU}(V) \cong \mathrm{SU}(2)$ up to overall phase [28]. Find a braid word w and a phase $z \in \mathrm{U}(1)$ such that

$$\|z\rho(w) - H\|_{\mathrm{op}} < 10^{-3}, \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Contrast this with the phase-only $\mathbb{Z}/3$ model.

2 Geometry

2.1 Differential forms and cohomology

A **smooth manifold** is a topological space equipped with an atlas whose transition maps are smooth. A smooth map $f : M \to N$ has a tangent map $df : TM \to TN$. The **tangent bundle** TM is the union of the tangent spaces, the **cotangent bundle** is T^*M, and a **vector field** is a smooth section of TM.

For an n-dimensional real vector space V, its **determinant line** is $\det(V^*) = \Lambda^n V^*$. An **orientation** of V is a choice of one of the two connected components of $\det(V^*) \setminus \{0\}$; the elements of the chosen component are positive. An orientation of an n-manifold M is a smoothly varying choice of orientation of its tangent spaces.

The exterior powers $\Lambda^k T_p^* M$ are those defined in the Algebra section. To define integration in a coordinate-free way, we will integrate differential forms over chains. The space of **differential n-forms** is $\Omega^n(M) = \Gamma(\Lambda^n T^*M)$. On an oriented n-manifold, a **volume form** is a nowhere-vanishing element of

$\Omega^n(M)$ that is positive with respect to the orientation. The exterior derivative $d : \Omega^n(M) \to \Omega^{n+1}(M)$ extends the differential of smooth functions and satisfies

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$. A **singular n-simplex** is a smooth map $\sigma : \Delta^n \to M$ where Δ^n is the standard n-simplex. The collection of singular n-simplices in M freely generates the abelian group $C_n(M)$ of **singular n-chains** in M. The **boundary map** $\partial : C_n(M) \to C_{n-1}(M)$ is defined on each simplex $\sigma : \Delta^n \to M$ by

$$\partial\sigma = \sum_{j=0}^{n} (-1)^j \sigma \circ f_j,$$

where $f_j : \Delta^{n-1} \to \Delta^n$ is the inclusion of the face opposite the j-th vertex of Δ^n.

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups A_n and homomorphisms $d_n : A_n \to A_{n-1}$ such that $d_{n-1} \circ d_n = 0$ for all n. The n-th **homology group** H_n of the chain complex is

$$H_n = \ker(d_n)/\operatorname{im}(d_{n+1}).$$

The prototypical example is singular homology which corresponds to the chain complex of singular chains with the boundary map defined above.

Cohomology and cochains are defined similarly, using superscripts instead of subscripts and boundary maps $d^n : A^n \to A^{n+1}$ increase degree instead of decrease it. Singular cohomology corresponds to the cochain complex given by dualizing the singular chain complex above.

The **de Rham cohomology** of M is

$$H^n_{\mathrm{dR}}(M) = \ker(d\colon \Omega^n(M) \to \Omega^{n+1}(M))/\operatorname{im}(d\colon \Omega^{n-1}(M) \to \Omega^n(M)).$$

For a smooth map $f : N \to M$, the **pullback** of $\alpha \in \Omega^p(M)$ is

$$(f^*\alpha)_x(v_1, \ldots, v_p) = \alpha_{f(x)}(df_x v_1, \ldots, df_x v_p).$$

Exercise 29 (de Rham theorem)

Prove that integration over smooth singular cycles induces a natural isomorphism between de Rham cohomology and singular cohomology with real coefficients.

Exercise 30 (Nontrivial cohomology produces the Aharonov–Bohm phase)

Outside an idealized solenoid, let

$$M = \mathbb{R}^3 \setminus \{(0,0,z) : z \in \mathbb{R}\}, \qquad d\theta = \frac{-y\,dx + x\,dy}{x^2 + y^2}, \qquad A = \frac{\Phi}{2\pi} d\theta.$$

For a particle of charge q, use de Rham cohomology to derive the gauge-invariant phase

$$\exp\left(\frac{iq}{\hbar}\int_\gamma A\right)$$

and the Aharonov–Bohm interference shift $q\Phi/\hbar$ [25].

2.2 Hodge theory and electromagnetism

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric section $g \in \Gamma(T^*M \otimes T^*M)$. On an oriented pseudo-Riemannian n-manifold, the metric induces a bilinear form $\langle \cdot, \cdot \rangle_g$ on differential forms. A **metric volume form** vol_g is a positive volume form satisfying

$$|\langle \mathrm{vol}_g, \mathrm{vol}_g \rangle_g| = 1.$$

The **Hodge star** is an operator $\star : \Omega^p(M) \to \Omega^{n-p}(M)$ satisfying

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \, \mathrm{vol}_g \, .$$

Exercise 31 (Metric volume and Hodge star)
Prove that an oriented pseudo-Riemannian manifold has a unique metric volume form and a unique Hodge-star operator. Determine $\star^2$ in terms of the degree and signature of the metric.

On an oriented Lorentzian four-manifold, the electromagnetic field strength is a two-form F and the electric four-current is a three-form J.

Exercise 32 (Maxwell's equations imply charge conservation)
From Maxwell's equations

$$dF = 0, \qquad d \star F = J.$$

Derive charge conservation in differential and integral form. On Minkowski spacetime, write

$$F = B - dt \wedge E, \qquad J = \rho \, \mathrm{vol}_3 - dt \wedge \star_3 j,$$

and recover the classical Maxwell equations and continuity equation.

Exercise 33 (Gauge invariance forces charge conservation)
For $F = dA$, consider

$$S[A] = -\frac{1}{2} \int_M F \wedge \star F + \int_M A \wedge J.$$

Derive Maxwell's equations and prove that gauge invariance is equivalent to charge conservation, assuming vanishing boundary terms.

2.3 Vector bundles and connections

A **smooth vector bundle** $p : E \to M$ is a smooth family of vector spaces locally isomorphic to a product $U \times V \to U$. For a smooth map $f : N \to M$, its **pullback bundle** is

$$f^*E = \{(x, e) \in N \times E : f(x) = p(e)\} \longrightarrow N.$$

Write $\Omega^0(M,E)=\Gamma(E)$. A **connection** on E is a ground-field linear map

$$\nabla:\Omega^0(M,E)\longrightarrow\Omega^1(M,E)$$

satisfying $\nabla(fs)=df\otimes s+f\nabla s$. Extend it to E-valued forms by, for $\alpha\in\Omega^p(M)$,

$$\nabla(\alpha\wedge\eta)=d\alpha\wedge\eta+(-1)^p\alpha\wedge\nabla\eta,$$

and define its **curvature** by $F_\nabla=\nabla^2\in\Omega^2(M,\operatorname{End}E)$. Equip $\operatorname{End}E$-valued forms with the induced connection.

Exercise 34 (Bianchi identity)
Prove that every vector-bundle connection satisfies $\nabla F_\nabla=0$.

2.4 K-theory

For a compact Hausdorff space X, let $\mathrm{Vect}(X)$ be the category of finite-rank complex vector bundles with continuous local trivializations over X, with the usual direct sum, tensor product, dual, and pullback constructions. The **Grothendieck group** $K^0(X)$ is generated by isomorphism classes $[E]$ subject to

$$[E\oplus F]=[E]+[F].$$

Define multiplication by $[E][F]=[E\otimes F]$. For a continuous map $f:X\to Y$ between compact Hausdorff spaces, define pullback by

$$f^*:K^0(Y)\longrightarrow K^0(X),\qquad [E]\longmapsto[f^*E].$$

Exercise 35 (Grothendieck group of vector bundles)
Prove that $(f\circ g)^*E\cong g^*f^*E$ *and that pullback preserves direct sums, tensor products, and duals. Let* X *and* Y *be compact Hausdorff spaces and let* $f:X\to Y$ *be continuous. Prove that direct sum and tensor product induce a commutative ring structure on* $K^0(X)$ *and that* f^* *is a ring homomorphism. Prove that*

$$K^0(\mathrm{pt})\cong\mathbb{Z}$$

via the rank map.

Fix a basepoint $x_0\in X$, regarded as a map $x_0:\mathrm{pt}\to X$. The **reduced K-theory** of the pointed space (X,x_0) is

$$\widetilde{K}^0(X)=\ker\left(K^0(X)\xrightarrow{x_0^*}K^0(\mathrm{pt})\right).$$

For a bundle E of rank r, its **reduced class** is

$$[E]-[\underline{\mathbb{C}}^r],$$

where $\underline{\mathbb{C}}^r=X\times\mathbb{C}^r$ is the trivial rank-r bundle. Write $K^n(X)$ for the generalized cohomology groups extending $K^0(X)$.

Exercise 36 (Stable equivalence and Bott periodicity)

Let E and F be complex vector bundles over a compact Hausdorff space X. Prove that $[E] = [F]$ in $K^0(X)$ if and only if there is a vector bundle G such that

$$E \oplus G \cong F \oplus G.$$

Deduce that the reduced class of E is unchanged after replacing E by $E \oplus \underline{\mathbb{C}}^n$. Prove Bott periodicity

$$K^{n+2}(X) \cong K^n(X)$$

and deduce that complex K-theory is naturally $\mathbb{Z}/2$-graded.

For $0 \leq r \leq N$, identify the complex Grassmannian $\mathrm{Gr}_r(\mathbb{C}^N)$ with the space of rank-r orthogonal projections in $M_N(\mathbb{C})$. Its **tautological bundle** has fiber $\operatorname{im} P$ over P. Let BU denote the stable complex Grassmannian obtained from finite Grassmannians by adjoining trivial summands.

Exercise 37 (Stable classification of complex vector bundles)

Classify stable complex vector bundles over a connected finite CW complex X by projection-valued maps, proving

$$K^0(X) \cong [X, \mathbb{Z} \times BU], \qquad \widetilde{K}^0(X) \cong [X, BU]_*.$$

Exercise 38 (Künneth theorem computes torus K-theory)

For finite CW complexes X and Y with torsion-free K-groups, prove that external product induces the graded isomorphism [1]

$$K^*(X) \mathbin{\widehat{\otimes}} K^*(Y) \cong K^*(X \times Y).$$

Here $\widehat{\otimes}$ denotes graded tensor product. Use Bott periodicity to deduce

$$K^*(\mathbb{T}^d) \cong \Lambda_{\mathbb{Z}}(\eta_1, \ldots, \eta_d), \qquad |\eta_j| = 1.$$

2.5 Chern classes and the Chern character

Let P be an invariant symmetric k-linear polynomial on the endomorphism algebra of a fiber of E, where invariant means invariant under simultaneous conjugation. Define

$$P(F_\nabla) = P(F_\nabla, \ldots, F_\nabla) \in \Omega^{2k}(M).$$

Exercise 39 (Chern–Weil theory produces Chern classes)

Prove that $P(F_\nabla)$ is closed and that its de Rham class is independent of the connection. Apply this to

$$\det\left(1 + \frac{iF_\nabla}{2\pi}\right) = 1 + c_1(E, \nabla) + c_2(E, \nabla) + \cdots$$

to define the Chern classes of E [19].

Exercise 40 (Chern character)

Define

$$\mathrm{ch}(E, \nabla) = \operatorname{tr} \exp\left(\frac{iF_\nabla}{2\pi}\right).$$

For compact M, prove that it induces a ring homomorphism

$$\mathrm{ch} : K^0(M) \longrightarrow H^{\mathrm{even}}_{\mathrm{dR}}(M).$$

2.6 Band topology

For the gapped families below, assume that X is connected, so vector-bundle ranks are constant. Let V be a finite-dimensional Hermitian vector space. Exercise 8 supplies the spectral decomposition used below. A continuous family is a map

$$H : X \longrightarrow \mathrm{End}(V)$$

and write $H_x = H(x)$. It is **gapped** if every H_x is self-adjoint and invertible. Given a positively oriented contour Γ enclosing the negative spectra and excluding the positive spectra, define

$$P_x^- = \frac{1}{2\pi i}\int_\Gamma (z - H_x)^{-1}\, dz, \qquad E_x^- = \mathrm{im}(P_x^-).$$

Exercise 41 (Gapped families define stable K-theory classes)

Show that the negative spectral subspaces of a gapped family form a vector bundle and that its class in $K^0(X)$ is invariant under gap-preserving homotopy. Show that

$$[E_H^-] - [\underline{\mathbb{C}}^{\,\mathrm{rank}\, E_H^-}] \in \widetilde{K}^0(X)$$

is invariant under adjoining constant gapped summands, and that every reduced class is represented by a flattened family $H_x = 1 - 2P_x$.

Exercise 42 (Berry phase as holonomy)

Let $H : X \to \mathrm{End}(V)$ be a smooth self-adjoint family with a simple isolated eigenvalue λ_x, and let $L \to X$ be its eigenline bundle. For a local unit eigenvector ψ, define

$$\mathcal{A} = i\langle \psi, d\psi\rangle, \qquad \mathcal{F} = d\mathcal{A}.$$

Prove that these local forms define a connection and curvature on L and that the Berry phase around a closed curve γ is its holonomy $\exp(i\int_\gamma \mathcal{A})$.

Exercise 43 (K-theory classifies class A band insulators)

A class A band model over a compact connected parameter space X has a smooth gapped Hamiltonian

$$H : X \longrightarrow \mathrm{End}(V)$$

whose negative-energy bundle consists of occupied states; for a crystal, $X = \mathbb{T}^d$. Prove that stable equivalence classes are classified by $\widetilde{K}^0(X)$ [2], and compute the strong phases

$$\widetilde{K}^0(S^d) \cong \begin{cases} \mathbb{Z}, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

Use Exercise 38 to compute the classifications over $\mathbb{T}^d$ for $d = 1, 2, 3$ and distinguish strong from weak invariants. For $d = 2$, use Exercise 39 to identify the integer invariant with the first Chern number

$$C_1(E_H^-) = \frac{i}{2\pi} \int_{\mathbb{T}^2} \operatorname{tr}\big(P^-\, dP^- \wedge dP^-\big),$$

and prove that a nonzero value obstructs a global smooth frame of occupied states.

2.7 Symplectic geometry

For a vector field X, contraction is the map $\iota_X : \Omega^p(M) \to \Omega^{p-1}(M)$ defined by

$$(\iota_X \alpha)(X_2, \ldots, X_p) = \alpha(X, X_2, \ldots, X_p).$$

Define the Lie derivative of forms by Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$.

A **symplectic form** is a two-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and $X \mapsto \iota_X \omega$ is a bundle isomorphism $TM \to T^*M$. Write $C^\infty(M)$ for the smooth real-valued functions on M. For $H \in C^\infty(M)$, define X_H by

$$\iota_{X_H} \omega = dH.$$

Let $\pi : T^*Q \to Q$ be the cotangent bundle of a configuration manifold. Its **tautological one-form** is

$$\theta_{(q,p)}(\xi) = p\big(d\pi_{(q,p)} \xi\big),$$

and its **canonical symplectic form** is $\omega_0 = -d\theta$.

Exercise 44 (The symplectic form produces Hamilton's equations)
For

$$H(q, p) = \frac{1}{2m} \sum_{i=1}^{n} p_i^2 + V(q),$$

derive Hamilton's equations intrinsically from ω_0, recover Newton's equation, and solve the harmonic-oscillator flow.

Exercise 45 (Magnetic symplectic curvature produces the Lorentz force)

For a particle of charge e in a closed magnetic field $B \in \Omega^2(\mathbb{R}^3)$, derive the Lorentz force from

$$\omega_B = \omega_0 - e\pi^* B \qquad \text{and} \qquad H(q, p) = \frac{|p|^2}{2m} + e\phi(q),$$

and relate the gauge freedom for $B = dA$ to the Aharonov–Bohm potential.

Exercise 46 (Liouville's theorem)
Prove that every Hamiltonian flow preserves the symplectic volume $\omega^n/n!$.

Exercise 47 (Hamiltonian phase volume produces ideal-gas thermodynamics)

For Liouville measure $d\mu_L = \omega^n/n!$, *show that Hamiltonian evolution preserves the fine-grained Gibbs entropy*

$$S_G[\rho] = -k_B \int_M \rho \log \rho \, d\mu_L.$$

For N *indistinguishable noninteracting particles in a box, divide phase volume by* $N!h^{3N}$ *and use*

$$H(q,p) = \sum_{a=1}^{N} \frac{|p_a|^2}{2m}.$$

Derive the Sackur–Tetrode entropy and the ideal-gas laws

$$E = \frac{3}{2} N k_B T, \qquad PV = N k_B T.$$

Derive the adiabatic and sound-speed laws, including, for $\varrho = mN/V$,

$$c = \sqrt{\frac{5P}{3\varrho}}.$$

Reconcile the entropy increase in free expansion with conservation of fine-grained Gibbs entropy.

For $f, g \in C^\infty(M)$, define

$$\{f, g\} = \omega(X_f, X_g).$$

Exercise 48 (Hamiltonian symmetry produces conservation laws)
Prove that a Hamiltonian f *generates a symmetry of* H *exactly when* f *is conserved by the flow of* H, *and distinguish this statement from Noether's variational theorem [17].*

Exercise 49 (Central forces and angular momentum)
On $T^*\mathbb{R}^3$, *let*

$$H(q,p) = \frac{|p|^2}{2m} + V(|q|), \qquad L = q \times p.$$

Derive the angular-momentum Poisson algebra and show that rotational symmetry confines each nonradial trajectory to a fixed plane.

2.8 Riemannian geometry

An **affine connection** is a connection on TM; write $\nabla_X Y = (\nabla Y)(X)$. For a covector field α, set

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

Define the covariant derivative of general tensor fields by the product rule and by requiring it to commute with contractions.

An affine connection is **torsion-free and metric-compatible** when

$$\nabla_X Y - \nabla_Y X = [X, Y], \qquad \nabla g = 0.$$

Such a connection is called a **Levi-Civita connection**.

Exercise 50 (Levi-Civita connection)

Prove that every pseudo-Riemannian manifold admits a unique Levi-Civita connection. Derive the coordinate-free Koszul formula.

An affinely parametrized curve $\gamma : I \to M$ is a **geodesic** if $\nabla_{\dot\gamma}\dot\gamma = 0$. The Lie derivative of the metric is

$$(\mathcal{L}_X g)(Y, Z) = X(g(Y, Z)) - g([X, Y], Z) - g(Y, [X, Z]),$$

and X is a **Killing vector field** if $\mathcal{L}_X g = 0$.

Let (M, g) be a pseudo-Riemannian manifold with Levi-Civita connection ∇.

Exercise 51 (Killing fields and conserved quantities)

Let X be a Killing vector field and let γ be an affinely parametrized geodesic. Prove that

$$\frac{d}{dt} g(\dot\gamma, \dot\gamma) = 0, \qquad \frac{d}{dt} g(X, \dot\gamma) = 0.$$

Deduce that $g(X, \dot\gamma)$ is constant along γ.

The **curvature** of ∇ is the $\mathrm{End}(TM)$-valued 2-form F_∇ determined by

$$R(X, Y)Z := F_\nabla(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

Exercise 52 (Curvature as a tensor)

For $f \in C^\infty(M)$ and vector fields X, Y, Z, prove that

$$R(fX, Y)Z = fR(X, Y)Z, \qquad R(X, fY)Z = fR(X, Y)Z,$$

$$R(X, Y)(fZ) = fR(X, Y)Z.$$

Prove that $R(X, Y) = -R(Y, X)$ and deduce that

$$R \in \Gamma\left(\Lambda^2 T^* M \otimes T^* M \otimes TM\right).$$

Exercise 53 (Geodesic deviation) *Let $\Gamma : (-\varepsilon, \varepsilon) \times I \to M$ be a smooth variation through affinely parametrized geodesics, and define the vector fields along Γ by*

$$T = \frac{\partial \Gamma}{\partial t}, \qquad J = \frac{\partial \Gamma}{\partial s}.$$

Prove that $[T, J] = 0$ and that

$$\nabla_T \nabla_T J = R(T, J)T.$$

Exercise 54 (The Lefschetz number is a categorical trace)

Show that bounded complexes of finite-dimensional vector spaces form a symmetric monoidal category and that the Lefschetz number defines a categorical trace.

Exercise 55 (Gauss–Bonnet theorem)

Let (M, g) be a compact oriented Riemannian surface with piecewise smooth boundary. With Gaussian curvature K, signed geodesic curvature k_g, and signed exterior angles θ_j, prove [18]

$$\int_M K \operatorname{vol}_g + \int_{\partial M} k_g \, ds + \sum_j \theta_j = 2\pi\chi(M).$$

Derive its consequences for round spheres, positively curved tori, and closed hyperbolic surfaces.

The trace used below is the ordinary trace defined in the Algebra section. Define the **Ricci tensor**, **scalar curvature**, and **Einstein tensor** by

$$\operatorname{Ric}(X, Y) = \operatorname{tr}\big(Z \mapsto R(Z, X)Y\big), \qquad S = \operatorname{tr}_g(\operatorname{Ric}), \qquad G = \operatorname{Ric} - \frac{1}{2} S g.$$

For a symmetric covariant two-tensor A, its **divergence** is the covector field

$$(\operatorname{div} A)(Y) = \operatorname{tr}_g\big((X, Z) \mapsto (\nabla_X A)(Z, Y)\big).$$

Exercise 56 (Contracted Bianchi identity and conservation)

Assume the second Bianchi identity

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Contract this identity to prove that

$$\operatorname{div} G = 0.$$

Deduce that if T is a symmetric covariant two-tensor satisfying

$$G + \Lambda g = \kappa T$$

for constants Λ, κ with $\kappa \neq 0$, then $\operatorname{div} T = 0$.

2.9 Yang–Mills theory and the Standard Model

Let G be a Lie group with Lie algebra $\mathfrak{g}$. A **principal G-bundle** is a smooth free right G-manifold P with quotient $M = P/G$ such that the projection $P \to M$ is a locally trivial fiber bundle. For a finite-dimensional representation $\rho : G \to \mathrm{GL}(V)$, the **associated bundle** is

$$E = P \times_G V = (P \times V)/\sim, \qquad (p \cdot g, v) \sim (p, \rho(g)^{-1} v).$$

A **principal connection** is a $\mathfrak{g}$-valued one-form A on P that is G-equivariant and reproduces the generators of the fundamental vector fields. In a local trivialization it is represented by an ordinary form $A \in \Omega^1(U, \mathfrak{g})$, with curvature

$$F_A = dA + A \wedge A \in \Omega^2(U, \mathfrak{g}).$$

Exercise 57 (Isospin motivates Yang–Mills connections)

Treat the proton and neutron as a nucleon doublet carrying the defining representation of $\mathrm{SU}(2)$*. Gauge this symmetry and derive*

$$A \longmapsto A^g = g^{-1} A g + g^{-1} dg.$$

Show that the resulting connection differentiates associated-bundle fields and has covariant curvature, and relate this construction to [20].

Exercise 58 (Yang–Mills action)

On an oriented Riemannian four-manifold, consider the Yang–Mills action

$$S(A) = \int_M \mathrm{Tr}(F_A \wedge \star F_A).$$

Prove gauge invariance, derive

$$d_A \star F_A = 0,$$

and recover source-free Maxwell theory in the abelian case.

Exercise 59 (Standard Model representations and symmetry breaking)

Take the Standard Model gauge group to be

$$G_{\mathrm{SM}} = \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y.$$

Write $(\mathbf{n}, \mathbf{m})_Y$ *for the representation of dimension* n *under* $\mathrm{SU}(3)_C$*, dimension* m *under* $\mathrm{SU}(2)_L$*, and weak hypercharge* Y*. Take one generation of Weyl fermion multiplets to be*

$$Q_L = (\mathbf{3}, \mathbf{2})_{1/6}, \qquad u_R = (\mathbf{3}, \mathbf{1})_{2/3}, \qquad d_R = (\mathbf{3}, \mathbf{1})_{-1/3},$$

$$L_L = (\mathbf{1}, \mathbf{2})_{-1/2}, \qquad e_R = (\mathbf{1}, \mathbf{1})_{-1},$$

and define $Q = T_3 + Y$. Let ϕ be a section of the Higgs bundle associated to $(\mathbf{1}, \mathbf{2})_{1/2}$, with

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \qquad D\phi = d\phi + \tfrac{i}{2}g\, W^a\sigma_a\phi + \tfrac{i}{2}g' B\phi,$$

where $\mu^2, \lambda > 0$ and g, g' are coupling constants. From these data derive the fermion charges, the symmetry breaking $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\mathrm{EM}}$, the W and Z masses, the massless photon, and cancellation of the hypercharge anomalies.

2.10 Chern–Simons theory and TQFT

Exercise 60 (Chern–Simons transgression)
For two connections on E, set

$$\nabla^t = (1-t)\nabla^0 + t\nabla^1, \qquad A = \nabla^1 - \nabla^0, \qquad F_t = F_{\nabla^t},$$

where $A \in \Omega^1(M, \operatorname{End} E)$, and define

$$\mathrm{CS}_P(\nabla^0, \nabla^1) = k\int_0^1 P(A, F_t, \dots, F_t)\, dt.$$

Prove that

$$P(F_{\nabla^1}) - P(F_{\nabla^0}) = d\ \mathrm{CS}_P(\nabla^0, \nabla^1).$$

Let $\mathfrak{g} \subseteq M_N(\mathbb{C})$ be a matrix Lie algebra whose trace pairing is nondegenerate. For $A \in \Omega^1(M, \mathfrak{g})$, wedge products include matrix multiplication. As above, write

$$F_A = dA + A \wedge A$$

and define

$$\mathrm{CS}(A) = \mathrm{tr}\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right).$$

Exercise 61 (Chern–Simons form)
Prove that

$$d\ \mathrm{CS}(A) = \mathrm{tr}(F_A \wedge F_A).$$

For a closed oriented three-manifold M, define

$$S_{\mathrm{CS}}(A) = \int_M \mathrm{CS}(A).$$

Exercise 62 (Chern–Simons equation)
For $\alpha \in \Omega^1(M, \mathfrak{g})$, set $A_t = A + t\alpha$. Prove that

$$\left.\frac{d}{dt}\right|_{t=0} S_{\mathrm{CS}}(A_t) = 2\int_M \mathrm{tr}(\alpha \wedge F_A).$$

Deduce that the critical points satisfy $F_A = 0$.

For $k \in \mathbb{R}$, set $S_k = (k/4\pi)S_{\mathrm{CS}}$.

Exercise 63 (Gauge transformation)

Let G be a compact, connected, simple, and simply connected matrix Lie group with Lie algebra $\mathfrak{g}$. For a smooth map $g : M \to G$, define

$$A^g = g^{-1}Ag + g^{-1}dg.$$

Normalize the invariant trace so that

$$\nu(g) = \frac{1}{24\pi^2}\int_M \operatorname{tr}\big((g^{-1}dg)^3\big) \in \mathbb{Z}.$$

Prove that

$$\mathrm{CS}(A^g) - \mathrm{CS}(A) = -d\,\operatorname{tr}(dg\,g^{-1} \wedge A) - \frac{1}{3}\operatorname{tr}\big((g^{-1}dg)^3\big).$$

Deduce that $\exp(iS_k)$ *is invariant under all gauge transformations when* $k \in \mathbb{Z}$.

Now restrict to $2+1$ spacetime dimensions. Let M be a closed oriented three-manifold and let $\ell > 0$, so the cosmological constant is $\Lambda = -\ell^{-2}$. After identifying the coframe representation with $\mathfrak{sl}(2,\mathbb{R})$, let

$$e, \omega \in \Omega^1(M, \mathfrak{sl}(2,\mathbb{R})), \qquad A^{\pm} = \omega \pm \frac{1}{\ell}e.$$

The pair (A^+, A^-) has gauge Lie algebra $\mathfrak{sl}(2,\mathbb{R}) \oplus \mathfrak{sl}(2,\mathbb{R})$. Write

$$d_\omega e = de + \omega \wedge e + e \wedge \omega, \qquad F_\omega = d\omega + \omega \wedge \omega.$$

Exercise 64 (Flat connections encode three-dimensional gravity)

Prove that $F_{A^+} = F_{A^-} = 0$ *is equivalent to*

$$d_\omega e = 0, \qquad F_\omega + \frac{1}{\ell^2} e \wedge e = 0.$$

Assuming that $e : TM \to \mathfrak{sl}(2,\mathbb{R})$ *is fiberwise invertible, identify these equations with the torsion-free condition and the Einstein equation with constant negative curvature.*

Exercise 65 (Three-dimensional gravity is a Chern–Simons theory)

For the first-order Einstein–Hilbert functional

$$S_{\mathrm{EH}}(e,\omega) = \int_M \operatorname{tr}\left(e \wedge F_\omega + \frac{1}{3\ell^2} e \wedge e \wedge e\right).$$

Prove that, up to normalization and an exact term, it equals

$$S_{\mathrm{CS}}(A^+) - S_{\mathrm{CS}}(A^-)$$

[23, 24].

For a connection A and a finite-dimensional representation V of its gauge group, parallel transport around an oriented closed curve γ gives the holonomy $\mathrm{Hol}_\gamma(A)$. Define the **Wilson loop**

$$W_V(\gamma, A) = \mathrm{tr}_V(\mathrm{Hol}_\gamma(A)) .$$

For a framed oriented link $L = \gamma_1 \sqcup \cdots \sqcup \gamma_r$ labeled by $V_1, \ldots, V_r$, set

$$W_L(A) = \prod_{j=1}^{r} W_{V_j}(\gamma_j, A).$$

For a framed link, define the formal Chern–Simons expectation value [21]

$$\langle W_L \rangle_k = \frac{\int e^{iS_k(A)} W_L(A)\, \mathcal{D}A}{\int e^{iS_k(A)}\, \mathcal{D}A}$$

Exercise 66 (Wilson loops and the Jones polynomial)

For normalized $SU(2)$ theory at level k, set

$$q = \exp\left(\frac{2\pi i}{k+2}\right).$$

Assume the normalized Wilson-loop expectations satisfy

$$q^{-1}\langle W_{L_+}\rangle_k - q\langle W_{L_-}\rangle_k = \left(q^{1/2} - q^{-1/2}\right)\langle W_{L_0}\rangle_k$$

and $\langle W_{\bigcirc}\rangle_k = 1$. Show that they determine the Jones polynomial and account for framing dependence.

Exercise 67 (Reshetikhin–Turaev TQFT)

Prove that a modular braided fusion category defines a Reshetikhin–Turaev TQFT

$$Z : \mathrm{Bord}_3^{\mathrm{or}} \longrightarrow \mathrm{Vect}_{\mathbb{C}} .$$

Here $\mathrm{Bord}_3^{\mathrm{or}}$ is the category of closed oriented surfaces and oriented three-dimensional bordisms. Relate its Wilson lines to the F- and R-matrices and, for $SU(2)_k$, to the Chern–Simons skein relation [22].

3 Analysis

3.1 Hilbert spaces and bounded operators

A **norm** on a complex vector space H is a function $\|\cdot\| : H \to [0, \infty)$ that is definite, homogeneous, and satisfies the triangle inequality. A normed space is a **Banach space** if every Cauchy sequence converges. A **Hilbert space** is a complex vector space with a Hermitian inner product $\langle \cdot, \cdot \rangle$ whose norm $\|x\| = \sqrt{\langle x, x\rangle}$ is complete. A linear map $A : H \to K$ is **bounded** if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty.$$

Write $\mathcal{B}(H,K)$ for the normed space of bounded operators and $\mathcal{B}(H) = \mathcal{B}(H,H)$.

The notions of adjoint, self-adjoint, normal, positive, and unitary extend from the finite-dimensional case to bounded operators on Hilbert spaces.

Exercise 68 (Cauchy–Schwarz inequality)

Prove that every inner-product space satisfies

$$|\langle x,y\rangle| \le \|x\|\|y\|.$$

Deduce the triangle inequality for the induced norm and prove that $\mathcal{B}(H,K)$ is complete when K is complete.

Exercise 69 (Adjoints of bounded operators)

Prove that every bounded operator between Hilbert spaces has a unique adjoint and establish the identities

$$(AB)^* = B^*A^*, \qquad \|A^*A\| = \|A\|^2.$$

3.2 Compact operators and the Fredholm alternative

A bounded operator $K : H \to L$ between Hilbert spaces is **compact** if the image of the closed unit ball has compact closure. Write $\mathcal{K}(H,L)$ for the compact operators and $\mathcal{K}(H) = \mathcal{K}(H,H)$.

Exercise 70 (Compact operators)

Give the sequential characterization of compact operators and prove that they form an adjoint-closed norm-closed operator ideal generated densely by finite-rank operators.

A bounded operator $T : H \to L$ is **Fredholm** if $\ker T$ and $L/\operatorname{im} T$ are finite dimensional and $\operatorname{im} T$ is closed. Its **Fredholm index** is

$$\operatorname{ind}(T) = \dim\ker T - \dim(L/\operatorname{im} T).$$

Exercise 71 (Fredholm alternative)

For $K \in \mathcal{K}(H)$ and $\lambda \neq 0$, prove the Fredholm alternative [3]: $\lambda 1 - K$ has index zero,

$$\dim\ker(\lambda 1 - K) = \dim\ker(\overline{\lambda} 1 - K^*) < \infty.$$

Characterize solvability of $(\lambda 1 - K)x = y$ and deduce that every nonzero spectral value of a compact operator is an eigenvalue of finite multiplicity, with zero the only possible accumulation point.

3.3 Quantum states, uncertainty, and dynamics

For $A \in \mathcal{B}(H)$, define the operator exponential by

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$

Exercise 72 (Bounded-operator exponentials)
Prove that the exponential series converges in operator norm and that

$$\frac{d}{dt}e^{tA} = Ae^{tA} = e^{tA}A.$$

Prove the semigroup law and the identity $e^{A+B} = e^A e^B$ *for commuting* A *and* B.

A finite-dimensional **quantum state** is a positive operator ρ with $\operatorname{tr}(\rho) = 1$. A unit vector ψ defines the pure state $\rho_\psi = |\psi\rangle\langle\psi|$. For a self-adjoint observable A, define

$$\langle A\rangle_\rho = \operatorname{tr}(\rho A), \qquad \operatorname{Var}_\rho(A) = \operatorname{tr}\big(\rho(A - \langle A\rangle_\rho 1)^2\big)\,.$$

For $\rho = \rho_\psi$, use the subscripts ψ instead of ρ.

Exercise 73 (Robertson uncertainty inequality)
For self-adjoint $A, B \in \mathcal{B}(H)$ *and a unit vector* ψ, *prove [13]*

$$\operatorname{Var}_\psi(A)\operatorname{Var}_\psi(B) \geq \frac{1}{4}\left|\langle\psi, [A,B]\psi\rangle\right|^2, \qquad [A,B] = AB - BA.$$

Determine the equality condition.

Exercise 74 (Commuting observables are conserved)
Let A, H *be self-adjoint and write* $U_t = e^{-itH}$. *Prove that* U_t *is unitary and that*

$$\frac{d}{dt}\langle U_t\psi, AU_t\psi\rangle = -i\langle U_t\psi, [H,A]U_t\psi\rangle.$$

Deduce that $[H, A] = 0$ *if and only if the expectation of* A *is constant under this evolution for every initial vector.*

Exercise 75 (No-cloning theorem)
Let $b \in H$ *be a unit vector and suppose that a unitary* $C : H \otimes H \to H \otimes H$ *satisfies* $C(v \otimes b) = v \otimes v$ *and* $C(w \otimes b) = w \otimes w$ *for unit vectors* v, w. *Prove that [12]*

$$\langle v, w\rangle = \langle v, w\rangle^2.$$

Deduce that no unitary map clones every unit vector.

3.4 Schmidt decomposition and entanglement

For finite-dimensional Hilbert spaces H_A, H_B, the **partial trace** over H_B is the linear map

$$\operatorname{Tr}_{H_B} : \operatorname{End}(H_A \otimes H_B) \longrightarrow \operatorname{End}(H_A)$$

characterized by

$$\operatorname{Tr}(A \operatorname{Tr}_{H_B}(X)) = \operatorname{Tr}((A \otimes 1_{H_B})X) \qquad (A \in \operatorname{End}(H_A)).$$

For finite-dimensional Hilbert spaces H_A, H_B and $\psi \in H_A \otimes H_B$, define

$$T_\psi : H_A^* \to H_B, \qquad T_\psi(\phi) = (\phi \otimes 1)(\psi).$$

A **Schmidt decomposition** of ψ is an expression

$$\psi = \sum_{i=1}^{r} s_i a_i \otimes b_i.$$

where (a_i) and (b_i) are orthonormal families and $s_i > 0$; r is the **Schmidt rank**. For a unit vector ψ, define the reduced density operators by

$$\rho_A = \mathrm{Tr}_{H_B}(|\psi\rangle\langle\psi|), \qquad \rho_B = \mathrm{Tr}_{H_A}(|\psi\rangle\langle\psi|).$$

Exercise 76 (Schmidt decomposition and entanglement)
Use singular-value decomposition to derive the Schmidt decomposition, characterize separability, and compute the entanglement entropy

$$S(\psi) = -\,\mathrm{Tr}(\rho_A \log \rho_A) = -\sum_i s_i^2 \log s_i^2,$$

where the s_i are the Schmidt coefficients.

3.5 CHSH, Gisin, and Tsirelson

Exercise 77 (Every entangled pure two-qubit state violates CHSH)
Let A_0, A_1 and B_0, B_1 be self-adjoint operators with eigenvalues in $\{-1, 1\}$, and define the CHSH operator

$$\mathcal{C} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Establish the classical CHSH bound 2. For the two-qubit state

$$\psi_\theta = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle, \qquad 0 \le \theta \le \frac{\pi}{4},$$

derive the maximal CHSH value [8]

$$2\sqrt{1 + \sin^2(2\theta)}.$$

Conclude that every entangled pure two-qubit state violates CHSH.

A **bipartite binary quantum correlation matrix** has entries

$$c_{ij} = \langle \psi, (A_i \otimes B_j)\psi \rangle,$$

where ψ is a unit vector. The operators A_i and B_j are self-adjoint unitaries on finite-dimensional Hilbert spaces.

For a real matrix $M = (M_{ij})$, define

$$\beta_{\mathrm{C}}(M) = \max_{\varepsilon_i, \delta_j \in \{-1,1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|$$

and

$$\beta_{\mathrm{Q}}(M) = \sup_{c} \left| \sum_{i,j} M_{ij} c_{ij} \right|,$$

where the supremum is over bipartite binary quantum correlation matrices.

Exercise 78 (Tsirelson's representation theorem and bound)
Prove Tsirelson's representation theorem [9]:

$$c_{ij} = \langle x_i, y_j \rangle.$$

Derive the sharp CHSH bound $2\sqrt{2}$.

3.6 Grothendieck inequality

The **real Grothendieck constant** $K_G^{\mathbb{R}}$ is the least constant for which

$$\beta_{\mathrm{Q}}(M) \le K_G^{\mathbb{R}} \beta_{\mathrm{C}}(M)$$

holds for every real matrix M.

Exercise 79 (Grothendieck's inequality)
Prove that $K_G^{\mathbb{R}} < \infty$ *[10, 11]. Use*

$$M_{\mathrm{CHSH}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

to prove $K_G^{\mathbb{R}} \ge \sqrt{2}$ *and interpret the inequality as a dimension-independent bound on quantum advantage.*

3.7 Von Neumann algebras and subfactors

For $\mathcal{S} \subseteq \mathcal{B}(H)$, its **commutant** is

$$\mathcal{S}' = \{T \in \mathcal{B}(H) : Tx = xT \text{ for every } x \in \mathcal{S}\}.$$

A **von Neumann algebra** is a unital $*$-subalgebra $M \subseteq \mathcal{B}(H)$ satisfying $M = M''$. A net (T_λ) converges to T in the **strong operator topology** if $T_\lambda \xi \to T\xi$ for every $\xi \in H$, and in the **weak operator topology** if

$$\langle T_\lambda \xi, \eta \rangle \longrightarrow \langle T\xi, \eta \rangle$$

for every $\xi, \eta \in H$.

Exercise 80 (Bicommutant theorem)
Let $A \subseteq \mathcal{B}(H)$ be a unital $*$-subalgebra. Prove that

$$A'' = \overline{A}^{\mathrm{SOT}} = \overline{A}^{\mathrm{WOT}}.$$

The **center** of a von Neumann algebra is $Z(M) = M \cap M'$, and M is a **factor** if $Z(M) = \mathbb{C}1$.

For a linear map $\Phi : M \to N$ between von Neumann algebras, its nth **matrix amplification** is

$$\Phi_n([x_{ij}]) = [\Phi(x_{ij})].$$

The map Φ is **completely positive** if every Φ_n is positive, and it is **normal** if it preserves suprema of bounded increasing nets of positive elements. A **normal $*$-representation** of M on a Hilbert space K is a normal unital $*$-homomorphism $\pi : M \to \mathcal{B}(K)$. A **Heisenberg-picture quantum channel** is a normal unital completely positive map between von Neumann algebras.

Exercise 81 (Normal CP maps have Stinespring dilations)
Let M be a von Neumann algebra and let $\Phi : M \to \mathcal{B}(H)$ be normal and completely positive. Prove Stinespring's theorem [14]: there are a Hilbert space K, a normal representation $\pi : M \to \mathcal{B}(K)$, and $V : H \to K$ such that

$$\Phi(x) = V^*\pi(x)V, \qquad V^*V = \Phi(1).$$

Deduce that every normal channel on $\mathcal{B}(H_0)$ has the environmental form

$$\Phi(x) = V^*(x \otimes 1_E)V$$

and, in finite dimensions, the Kraus and partial-trace forms

$$\Phi_*(\rho) = \sum_\alpha K_\alpha \rho K_\alpha^* = \mathrm{Tr}_E(V\rho V^*).$$

Conclude that no quantum channel clones every pure state.

A positive functional ψ is **normal** if it preserves suprema of bounded increasing nets of positive elements and **faithful** if $\psi(a^*a) = 0$ implies $a = 0$. A **type II_1 factor** is an infinite-dimensional factor with a faithful normal tracial state

$$\tau : M \longrightarrow \mathbb{C}, \qquad \tau(ab) = \tau(ba), \qquad \tau(1) = 1.$$

For projections $p, q \in M$, write $p \sim q$ if there is $v \in M$ such that

$$v^*v = p, \qquad vv^* = q.$$

Write $p \precsim q$ if p is equivalent to a projection r satisfying $rq = r$.

Exercise 82 (Projection dimension)

Let M be a type II_1 factor with normalized trace τ. Prove that

$$p \sim q \quad \Longrightarrow \quad \tau(p) = \tau(q),$$

and prove the comparison theorem

$$p \precsim q \quad \Longleftrightarrow \quad \tau(p) \leq \tau(q).$$

Deduce that $\tau(p) = \tau(q)$ if and only if $p \sim q$.

A von Neumann algebra is **finite** if $v^*v = 1$ implies $vv^* = 1$. Let $N \subseteq M$ be finite von Neumann algebras with a common normalized trace τ. A **trace-preserving conditional expectation** is a normal unital completely positive map $E_N : M \to N$ such that

$$\tau \circ E_N = \tau, \qquad E_N(n_1 x n_2) = n_1 E_N(x) n_2$$

for $x \in M$ and $n_1, n_2 \in N$. Let $L^2(M, \tau)$ be the completion of M for

$$\langle x, y \rangle_\tau = \tau(y^* x).$$

Let M act on $L^2(M, \tau)$ by left multiplication.

Exercise 83 (Conditional expectation and Jones projection)

Prove that there is a unique trace-preserving conditional expectation $E_N : M \to N$. Prove that E_N extends to the orthogonal projection

$$e_N : L^2(M, \tau) \longrightarrow L^2(N, \tau).$$

Prove that

$$e_N = e_N^* = e_N^2, \qquad e_N x e_N = E_N(x) e_N$$

for every $x \in M$.

The **basic construction** is

$$M_1 = \langle M, e_N \rangle'' \subseteq \mathcal{B}(L^2(M, \tau)),$$

the double commutant of the algebra generated by M and e_N. For a projection $p = [p_{ij}] \in M_k(N)$ acting by left multiplication, define

$$\dim_N\big(pL^2(N, \tau)^k\big) = \sum_{i=1}^{k} \tau(p_{ii}).$$

An inclusion $N \subseteq M$ has **finite Jones index** if $L^2(M, \tau)$ is isomorphic as a right N-module to such a module; then

$$[M : N] = \dim_N L^2(M, \tau).$$

Exercise 84 (Jones basic construction)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that M_1 is a finite factor and that its normalized trace satisfies [4]

$$\tau_1(xe_Ny) = \frac{1}{[M:N]}\tau(xy)$$

for $x, y \in M$.

The **Jones tower** is the sequence obtained by iterating the basic construction:

$$N = M_{-1} \subset M = M_0 \subset M_1 \subset M_2 \subset \cdots .$$

Let e_i be the Jones projection used to construct M_i from M_{i-1}, and set

$$\delta = \sqrt{[M:N]}.$$

The **Temperley–Lieb algebra** with loop parameter δ is generated by U_i subject to

$$U_i^2 = \delta U_i, \qquad U_iU_j = U_jU_i \quad (|i-j| \geq 2), \qquad U_iU_{i\pm1}U_i = U_i.$$

Exercise 85 (Temperley–Lieb relations)

Prove that the Jones projections satisfy [7, 5]

$$e_i^2 = e_i = e_i^*, \qquad e_ie_j = e_je_i \quad (|i-j| \geq 2),$$

$$e_ie_{i\pm1}e_i = \delta^{-2}e_i.$$

Deduce that $U_i \mapsto \delta e_i$ defines a representation of the Temperley–Lieb algebra.

Exercise 86 (Braid representation)

Let $q \in \mathbb{C}^\times$ satisfy $\delta = q + q^{-1}$ and define $\sigma_i = q\,1 - U_i$. Prove that σ_i is invertible with inverse $q^{-1}1 - U_i$, and prove that

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1},$$

$$\sigma_i\sigma_j = \sigma_j\sigma_i \qquad (|i-j| \geq 2).$$

Deduce that the σ_i define a representation of the braid group [6].

A **Markov trace** on the Temperley–Lieb tower is a compatible family of positive normalized traces satisfying

$$\mathrm{tr}(xU_i) = \delta^{-1}\,\mathrm{tr}(x)$$

whenever x belongs to the algebra generated by the preceding generators.

Exercise 87 (Positivity quantizes subfactor indices below four)
Define

$$P_0(\delta) = 1, \qquad P_1(\delta) = \delta, \qquad P_{n+1}(\delta) = \delta P_n(\delta) - P_{n-1}(\delta).$$

Prove that the normalized traces on the Jones tower restrict to a Markov trace. Prove that, when $\sin\theta \neq 0$*,*

$$P_n(2\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}.$$

Use positivity of the Markov trace to prove that if $0 < \delta < 2$*, then*

$$\delta = 2\cos\left(\frac{\pi}{m}\right)$$

for some integer $m \geq 3$*.*

Exercise 88 (Jones index theorem)
Let $N \subseteq M$ *be an inclusion of type* II_1 *factors of finite Jones index. Prove that [5]*

$$[M : N] \in \left\{4\cos^2\left(\frac{\pi}{n}\right) : n \geq 3\right\} \cup [4, \infty).$$

Relate the discrete values to the $SU(2)_k$ *braid representations and Chern–Simons Wilson-loop invariants constructed above.*

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